

Neil Gandhi

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EDUCATION

The Pennsylvania State University | Schreyer Honors College

University Park, PA

Bachelor of Science in Mechanical Engineering

Expected Fall 2027

- **GPA:** 3.67/4.0
- **Relevant Coursework:** Mechatronics, Mechanical Design, ME Design Methodology, Fluid Flow, Mechanical Vibrations, Manufacturing Processes, Engineering Materials, Circuits & Instrumentation, Thermodynamics I, Dynamics, Strength of Materials, ODE/PDE

EXPERIENCE

Pittsburgh Drone Services

Pittsburgh, PA

Engineering Intern

Summer 2026

- Merged three drone thermal GeoTIFFs into an orthomosaic and built a MATLAB pipeline computing surface statistics across a 34–156°F range (76°F mean, 4.1°F σ).
- Modeled photovoltaic efficiency loss from surface temperature: 0.27% average loss on residential roofs versus up to 17.5% on 156°F industrial roofs, directing siting toward elevated racking and reflective coatings.
- Applied Z-score filtering and land-cover binning to isolate extreme heat to 5.98% of surface area (87% moderate, 12.9% vegetated); authored the Larimer and Manchester reports in QGIS.

RESEARCH EXPERIENCE

Pangborn Advanced Controls Lab (PACLAB), Penn State

University Park, PA

Undergraduate Researcher, Schreyer Honors Thesis

Aug. 2026 – Present

- Designing and fabricating a 3D-printed drone airframe in carbon-fiber-reinforced filament as the subject of a Schreyer Honors thesis.
- Determining the governing design and performance parameters for the printed airframe, evaluating material and geometric trade-offs through analysis and testing.

Penn State Behrend

Erie, PA

Undergraduate Research Assistant

May 2025 – Jan. 2026

- Built a three-dimensional non-linear ODE model extending the Lotka-Volterra system to one predator (wolf spider) and two competing prey species.
- Applied Jacobian linearization, the Center Manifold Theorem, and trapping regions to locate equilibria, then used MATLAB to generate 3D phase-plane trajectories confirming populations remain bounded.
- **Manuscript:** Second author of six on “A one-predator, two-prey model, and a two-predator, one-prey model, inspired by Wolf spider (Lycosidae) populations,” proving the two systems are equivalent under time reversal.

PROJECTS

Autonomous Line-Following Vehicle | *Embedded Systems*

Nov. 2024 – Jan. 2025

- Designed and built an autonomous vehicle using an Arduino microcontroller and photoresistor arrays for closed-loop path tracking.
- Wrote C++ control algorithms to process continuous sensor input and adjust DC motor PWM signals for real-time trajectory correction.

HackPSU Competition | *Full-Stack Developer*

Mar. 2026

- Built and deployed a full-stack web application in a 24-hour sprint, owning the Python/Flask backend against a Next.js frontend.

LEADERSHIP & INVOLVEMENT

MATE UROV Competition | *Lead Mechanical Contributor & Mentor*

Aug. 2021 – Jun. 2025

- Directed mechanical design and systems integration for a neutrally buoyant Remotely Operated Vehicle, earning the Best Engineering Presentation award.
- Led two multidisciplinary teams of 15+ students, coordinating mechanical, electrical, and propulsion subsystems for the RANGER and SCOUT competition classes.

Penn State Behrend Pickleball Club | *Founder, President*

Jan. 2025 – Dec. 2025

- Founded a new student organization, growing membership to 40+ students across campus.

HONORS & AWARDS

Schreyer Honors College Scholar (2026 – Present) | Behrend Honors Program (2024 – 2025) | The President Walker Award (2024)

TECHNICAL SKILLS

CAD & Simulation: SolidWorks, Fusion 360, AutoCAD, Finite Element Analysis (FEA), Computational Fluid Dynamics (CFD), Additive Manufacturing (FDM 3D Printing), Carbon Fiber-Reinforced Filament

Programming & Tools: MATLAB/Simulink, Python, C++, LaTeX, QGIS, Arduino, Embedded Systems, PID & Closed-Loop Control, Flask, Next.js

AI Tooling: Claude Code, Model Context Protocol (MCP) servers & plugins, OpenAI Codex, LLM-assisted design and development

Organizations: Nittany Motorsports (Formula SAE), Chess Club